

Carbonation of one-part alkali-activated materials incorporated by MgO: Phase characteristics and micromechanical properties

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ABSTRACT

The role of magnesium oxide (MgO) in the carbonation of cementitious materials, particularly alkali-activated materials (AAM), which is a promising low-carbon alternative to ordinary Portland cement (OPC), has garnered significant attention. However, there remains a lack of clear understanding regarding its underlying mechanisms and interrelation with calcium oxide (CaO) content. This study aims to understand the influence of MgO on carbonation mechanisms, carbonation products, and (C, N)–A–S–H gel micromechanical properties in AAM. Four different binder formulations were examined by varying contents of slag, class F fly ash, and light-burned MgO. A powder-type sodium metasilicate was used as an alkaline activator. CO₂ curing was conducted in a CO₂ incubator with 1 % concentration and under atmospheric pressure for up to 56 days using pulverized paste binders. The carbonation mechanism, polymorphs of carbonation products, and the micromechanical properties of the (C, N)–A–S–H gel matrix were characterized by conducting laboratory tests including pH measurement, XRD, TGA, DSC, nitrogen adsorption, and nanoindentation at various carbonation stages. A noticeable pattern of pH initially dropping and then rising was observed. Integrated XRD-TGA demonstrated the evolution, polymorphism, and amount of calcium carbonate phases. Higher MgO content in AAM binders favored the formation of aragonite and hydrotalcite. Extensive nanoindentation results and their statistical analysis revealed binder-specific trends in the pozzolanic matrix stiffness before and after carbonation. The findings of this study can be used in developing AAM binders as building materials and to provide a deeper understating of the carbonation reactions in AAM binders.

1. Introduction

Alkali-activated material (AAM) is a promising alternative to the ordinary Portland cement (OPC) binders, with the potential to be a carbon-neutral construction material [1]. This study examines the carbonation reaction in AAM binders. Carbonation in cementitious materials occurs when carbon dioxide dissolves into the pore solution. This process leads to the acidification of the pore solution and the formation of carbonate and bicarbonate ions. As a result, the alkalinity of the pore solution decreases, triggering the leaching of calcium from the solid matrix [2]. In OPC binders, portlandite, Ca(OH)₂, is highly soluble and readily dissolves into the pore solution [3]. However, AAMs may have a deficiency of portlandite, leading to the direct dissolution of the solid matrix composed of calcium (or sodium) aluminosilicate hydrate gel, known as (C, N)–A–S–H gel. Thus, (C, N)–A–S–H gel potentially

provides the Ca²⁺ ion source for calcium carbonate precipitation. Therefore, carbonation can yield leaching calcium ions from (C, N)–A–S–H gel and its transformation into amorphous silica gel [2,4–7]. The role of magnesium oxide (MgO) in carbonation and carbon dioxide sequestration of cementitious materials, particularly AAM, has garnered significant attention among researchers. However, there remains a lack of clear understanding regarding its underlying mechanisms.

One hypothesis or theory, proposed by Bernal et al. [8], links the role of MgO in carbonation to the presence of Mg–Al layered double hydroxide (LDH) or hydrotalcite phases. These phases possess a structure characterized by positively charged octahedral brucite, Mg(OH)₂ layers, which occur due to the substitution of some of the original Mg²⁺ ions with Al³⁺ ions. To balance the excess positive charges, charge-balancing anionic molecules and water are incorporated [9,10]. The carbonate ions possess a high potential for incorporation into the hydrotalcite

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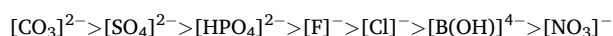
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structure due to the selective ionic exchange process. This exchange follows the order of ion exchange equilibrium constants outlined by Miyata [11]:



In alkali-activated slag (AAS) binders that incorporate slags with high MgO content (>5 %), the formation of hydrotalcite as a secondary product occurs. Shi et al. [12] suggested that the favorable formation of hydrotalcite-like phases is determined by the Mg/Al ratio of the slag, which falls within the range of 1.92–4.35. These hydrotalcite phases can function as CO₂ sorbents by capturing and exchanging carbonate ions, thus reducing CO₂ diffusion into AAS binders. Additionally, the formation of hydrotalcite leads to a lower uptake of aluminum in the C–A–S–H gel, as a portion of the available alumina is consumed in the formation of hydrotalcite [13]. Having a C–A–S–H gel with a reduced number of alumina bridging sites makes it less vulnerable to calcium leaching, which could potentially have a slight impact on decreasing the formation of calcium carbonate [14–16]. Apart from naturally occurring hydrotalcite, few studies have investigated the addition of synthesized hydrotalcite and its role in enhancing carbonation resistance. Shui et al. [17] examined the applications of calcined hydrotalcite with varying levels of calcination in the carbonation mechanism of Portland cement concrete. Zhang et al. [18] studied the addition of calcined hydrotalcite with different contents in AAS mortars. Lee et al. [19] examined alkali-activated slag/ash blends containing 0 %, 5 %, and 10 % reactive MgO under a 5 % CO₂ concentration. Their major findings were as follows: first, the addition of MgO did not result in the formation of magnesium silica hydrate (M–S–H) gel; second, TGA analysis indicated that the combined presence of MgO and carbonation favored the formation of hydrotalcite phase; and lastly, higher MgO content contributed to a decrease in the depth of pH drop, as indicated by phenolphthalein indicator.

Another theory, proposed by Morandau et al. [20], associates the role of MgO with the formation of stabilized amorphous calcium carbonate (ACC). They investigated AAS binders with 7 % and 13 % MgO content under accelerated carbonation using an aggressive 100 % CO₂ concentration. Their main finding was that binders with higher magnesium content tend to form Mg-stabilized ACC, which prevents the crystallization of calcium carbonate. This observation of magnesium's role in stabilizing ACC was also noted in synthetic calcium carbonate precipitation [21]. It was further reported that the ACC phase can regulate the decalcification of the C–A–S–H gel, as the solubility of ACC exceeds that of C–A–S–H gel at typical pH levels, thereby impeding calcium leaching from the gel. This was supported by a comparison of the Ca–Si correlation in the X-ray pair distribution functions (PDF) of AAS binders with 7 % and 13 % MgO content [20].

The MgO content not only hinders calcium carbonate precipitation but also influences the types of polymorphs that are formed. Studies on synthesized calcium carbonates have shown that the MgO content can govern the incorporation of MgO within the calcite lattice or induce aragonite precipitation, depending on the Mg/Ca ratio [21,22]. Additionally, high-magnesium calcite has higher solubility compared to pure calcite [21,23]. It is worth noting that various parameters such as CO₂ curing concentration and duration, alkaline activator type, and the age of the AAM at the start of carbonation can also impact the resulting calcium carbonate polymorphs [19,24]. Furthermore, the addition of MgO can potentially trigger the formation of M–S–H gel, although M–S–H gel has been reported to be stable only at pH values between 8 and 10.4 [25].

In this paper, we examine the effects of carbon dioxide curing with a 1 % concentration on one-part AAM binders, with and without the inclusion of light-burned MgO. The selection of MgO as an additive is based on its ability to change the carbonation reaction kinetics and products in AAM binders. Previous research on the role of MgO in AAM binders has reported diverse mechanisms and hypotheses. Therefore, it

is crucial to determine whether the reduced carbonation rate is primarily attributed to the formation of hydrotalcite or the stabilization of amorphous calcium carbonate or both. Furthermore, previous studies have presented the precipitation of different types of calcium carbonate polymorphs, creating a knowledge gap regarding the relationship between polymorphs and the MgO content. Additionally, the effect of MgO on the micromechanical properties of the (C, N)–A–S–H gel remains unknown, raising questions about whether it has a positive or negative impact.

In summary, this study aims to understand the influence of MgO additive on carbonation mechanisms, carbonation products, and (C, N)–A–S–H gel micromechanical properties. Two different AAM precursors, i.e., 100 % slag and 50 % slag-50 % class F fly ash are used as binder materials. The 100 % slag precursor has a higher calcium-to-silica ratio than the 50 % slag-50 % class F fly ash mixture. Calcium leaching of (C, N)–A–S–H gel with different calcium-to-silica ratios is of interest in cementitious materials research. Laboratory tests are conducted in multiple aspects incorporating physical, chemical, and mechanical analysis to assess carbonated and uncarbonated AAM binders. The experimental results are utilized to elucidate linkages between processing, properties, and microstructure characteristics of one-part AAMs over carbonation.

2. Experimental program

2.1. Materials

Ground granulated blast-furnace slag (GGBFS) and class F fly ash were selected as the aluminosilicate precursors. GGBFS (complying with ASTM C989 requirements [26]) and class F fly ash (complying with ASTM C618 requirements [27]) were obtained from Skyway Slag Cement LLC and Boral Resources LLC, respectively. Table 1 shows chemical characterization from X-ray fluorescence (XRF) test of the fly ash and slag. It is evident that the slag has a higher CaO content than the fly ash, with the CaO content in the fly ash being below 10 %. Moreover, the total of pozzolanic oxides (i.e., SiO₂ + Al₂O₃ + Fe₂O₃) is higher than 70 %, thus the fly ash can be categorized as class F. Fig. 1 shows the X-ray diffraction (XRD) patterns of GGBFS and fly ash. It can be seen that GGBFS is composed of approximately 99 % amorphous phase. Quantitative XRD analysis on fly ash is also displayed in Fig. 2. Results show that the fly ash is composed of around 79 % amorphous phase. Powdered sodium metasilicate was utilized as an alkaline activator with a SiO₂/Na₂O molar ratio of approximately 1.0 and a pH of 12.5 at room temperature. To enhance the reactive MgO content in the AAM binders, light-burned MgO with high purity was used as an additive. Light burned MgO is a reactive additive subjected to calcination within the temperature range of 700°C to 1000°C. Chemical analysis of the light burned MgO shows a composition of 98.2 % MgO. Physical analysis of the light burned MgO also shows an average particle size of 3–8 microns and a surface area of 20–30 m²/g.

2.2. Sample preparation

AAM binders were synthesized at paste level, following the guidelines outlined in ASTM C305. The process involved mixing the solid activators and precursors in a dry state for approximately 3 minutes at a low speed of 140 ± 5 rpm, followed by the gradual addition of water to the dry mix. The binder was then mixed for an additional 5 minutes at the same speed of 140 ± 5 rpm. The mixer was paused for two minutes, and the mixture was mixed by hand before being placed into the molds. As shown in Table 2, four groups of binder were investigated in this study. The AAS mix utilized 100 % slag, while the AASF mixture was composed of 50 % slag and 50 % class F fly ash. AASM binder utilized 100 % slag with 10 % MgO additive and AASFM binder was composed of 50 % slag and 50 % class F fly ash with 10 % MgO. It should be noted that the additive ratio is 10 % of the precursors' weight.

Table 1
XRF characterization of precursors.

Precursor	CaO Weight (%)	SiO ₂	Al ₂ O ₃	Fe ₂ O ₃	SO ₃	MgO	K ₂ O	Na ₂ O	LOI
Fly Ash	6.87	53.51	19.67	10.24	0.88	1.41	2.60	0.88	0.96
GGBFS	33.76	41.47	12.90	0.61	1.76	8.92	0.29	0.27	0.02

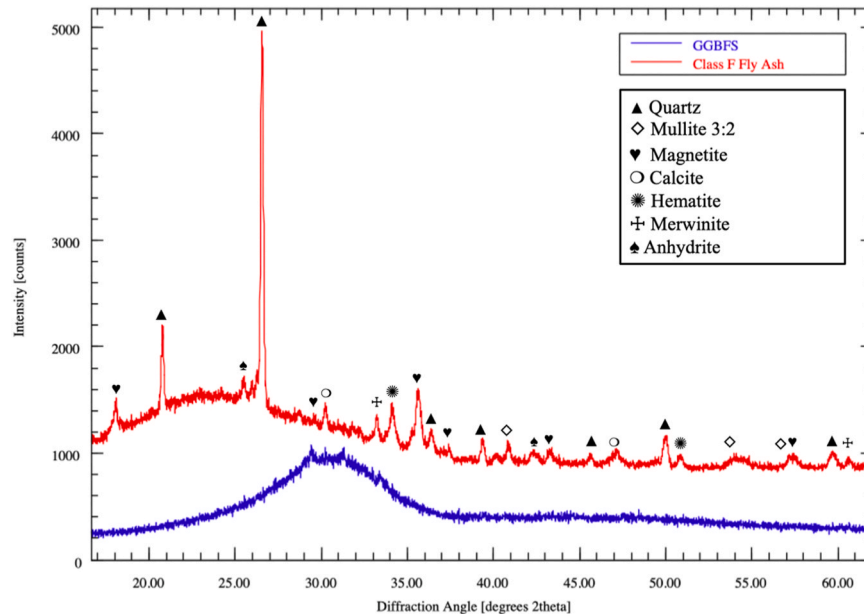


Fig. 1. XRD patterns of raw GGBFS and fly ash.

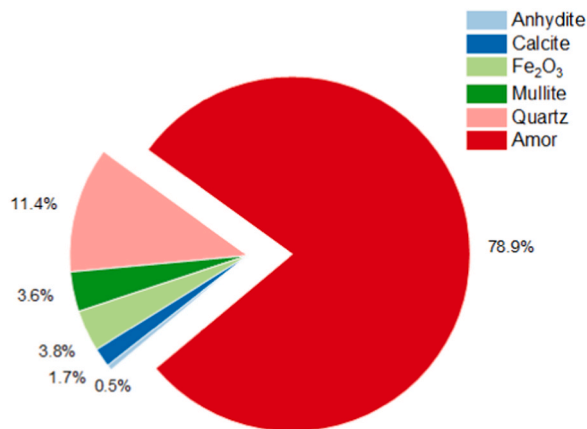


Fig. 2. Quantitative XRD analysis of fly ash.

To obtain a more analytical insight of the fabricated binders, the AAM binder proportions were translated into molar units. For this purpose, the weight percentages of reactive CaO, MgO, Al₂O₃, SiO₂, and equivalent Na₂O were determined from XRD and XRF analyses on raw

Table 2
AAM binder proportions.

ID	Precursor	Activator	Additive	Water to Precursor Ratio	Activator to Precursor Ratio	Additive to Precursor Ratio
AAS	100 % Slag	Sodium Metasilicate	N.A.	0.4	0.1	N.A.
AASF	50 % Slag- 50 % F Ash	Sodium Metasilicate	N.A.	0.35	0.15	N.A.
AASM	100 % Slag	Sodium Metasilicate	MgO	0.4	0.1	0.1
AASFM	50 % Slag- 50 % F Ash	Sodium Metasilicate	MgO	0.35	0.15	0.1

^a Note: N.A. stands for not available.

GGBFS and fly ash. The number of moles of each oxide in 100 g of AAM paste was then calculated and is presented in Table 3.

For example, in 100 g of AASM binder, the slag weight is approximately 62.5 g. From XRF analysis, the weight percentage of MgO in the slag is 8.92 %. In addition, in 100 g of AASM binder, 6.25 g of MgO additive with a purity of 98.2 % is used. As a result, the approximate weight of MgO in 100 g of AASM is 11.70 g. By dividing the MgO weight by the MgO molar weight (40.3 g/mol), the number of MgO moles can be calculated as 0.29 mol. Table 3 summarizes the calculations of different elemental oxides for 4 binders. Subsequently, the CaO/SiO₂ and MgO/SiO₂ molar ratios are calculated. According to this calculation,

Table 3
Elemental oxide analysis per 100 g of AAM binder.

Mix ID	MgO	CaO	Al ₂ O ₃	SiO ₂	Na ₂ O Equivalent ^a	CaO/ SiO ₂ Ratio	MgO/ SiO ₂ Ratio
	Mol.						
AAS	0.15	0.40	0.08	0.52	0.06	0.77	0.29
AASF	0.09	0.24	0.10	0.59	0.10	0.41	0.15
AASM	0.29	0.38	0.08	0.48	0.05	0.79	0.60
AASFM	0.23	0.23	0.09	0.50	0.09	0.46	0.46

^a Note: Na₂O Equivalent = Na₂O + 0.658 K₂O.

CaO/SiO₂ ratio for AAS and AASM binders is in a range of 0.77–0.79, while CaO/SiO₂ ratio for AASF and AASFM binders is in a range of 0.41–0.46. This can clearly highlight the difference of Ca/Si ratio between the two binders: 100 % slag and 50 % slag-50 % fly ash.

Paste specimens were cast into cylindrical silicon molds with a diameter of 25 mm (1 in.) and a height of 12.5 mm (0.5 in.). Initial curing was carried out for a duration of 28 days at a temperature of 24 ± 2 °C and a relative humidity (RH) of 95 ± 2 %. These conditions were chosen to maximize the hydration and to minimize the occurrence of carbonation. It is well-known that high moisture content can significantly slow down the carbonation process as the diffusion coefficient of CO₂ is significantly lower in the aqueous state compared to the gas state. After the initial curing, paste specimens were crushed using a mortar and pestle. Then, the pulverized paste specimens were filtered by sieving, where the collected specimens passed sieve No. 16 (1.18 mm) and retained on sieve No. 30 (600 µm). This sample preparation was conducted to achieve a homogeneous surface area and a uniform CO₂ ingress during the carbonation stage. An optical microscope image of the pulverized paste particles is shown in Fig. 3.

2.3. CO₂ curing procedure

In this study, CO₂ curing was conducted in an incubator under controlled conditions, with a CO₂ concentration of 1 ± 0.1 %, a temperature of 24 ± 2 °C, a relative humidity (RH) of 65 ± 2 %, and under atmospheric pressure. The selection of the CO₂ concentration aimed to achieve a low carbonate/calcium ratio in the pore solution. Each binder underwent curing for three different durations: 0, 28, and 56 days. It should be noted that the specimens undergoing 0-day CO₂ curing were carefully stored in a controlled chamber with a temperature maintained at 24 ± 2 °C and a relative humidity (RH) of 95 ± 2 %. Table 4 provides a list of the studied binders, categorized based on their composition and curing stage.

2.4. Summary of testing methods

To monitor the pH changes of AAM binders, pore solution of the AAM binders were extracted using the hot water-extraction method [28, 29]. The pulverized samples were further ground using a mortar and pestle to obtain a powder sample. The powder was sieved to achieve a particle size below 150 µm. This powder was mixed with deionized water at a ratio of 10 g to 100 ml. This suspension was boiled for around 10 minutes and left to stand for approximately 24 hours at room temperature. To obtain the pore solution, the suspension was filtered, and the volume of the solution was adjusted to 100 ml. The pH of pore solution was measured using a Thermo Fisher Scientific bench-top pH meter.

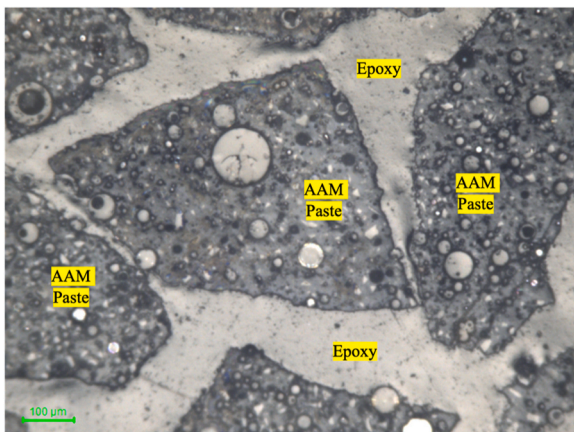


Fig. 3. Optical microscope image of pulverized AAM binder.

Table 4

List of binders categorized based on composition and curing stage.

Mixture ID	Mixture Type
AAS-C0	100 % slag – No CO ₂ curing
AAS-28C1	100 % slag – 1 % CO ₂ curing for 28 days
AAS-56C1	100 % slag – 1 % CO ₂ curing for 56 days
AASF-C0	50 % slag- 50 % F ash – No CO ₂ curing
AASF-28C1	50 % slag- 50 % F ash – 1 % CO ₂ curing for 28 days
AASF-56C1	50 % slag- 50 % F ash – 1 % CO ₂ curing for 56 days
AASM-C0	100 % slag with 10 % MgO Additive – No CO ₂ curing
AASM-28C1	100 % slag with 10 % MgO Additive – 1 % CO ₂ curing for 28 days
AASM-56C1	100 % slag with 10 % MgO Additive – 1 % CO ₂ curing for 56 days
AASFM-C0	50 % slag- 50 % F ash with 10 % MgO Additive – No CO ₂ curing
AASFM-28C1	50 % slag- 50 % F ash with 10 % MgO Additive – 1 % CO ₂ curing for 28 days
AASFM-56C1	50 % slag- 50 % F ash with 10 % MgO Additive – 1 % CO ₂ curing for 56 days

X-ray diffraction (XRD) test was utilized for chemical analysis of AAM binders. The testing was carried out using a Bruker D8 ENDEAVOUR with a 2θ range from 5° to 60° on powder samples. The powder samples were obtained similarly to the procedure used in the hot water extraction. Quantitative analysis was performed on the XRD patterns per the Rietveld refinement method to evaluate the metal carbonates polymorphs and phase changes before and after carbonation.

Thermal analysis was conducted using a TA Instruments Discovery SDT. The thermal analysis was composed of two different analytical methods: thermogravimetric analysis (TGA) and differential scanning calorimetry (DSC). The experiment design was performed using a 10 °C/min heating rate from 30 to 1000 °C. TGA was used to measure the changes in weight because of the thermal increment. DSC quantifies the variation in heat required to raise the temperature of a specimen relative to a reference material, in relation to temperature [30]. Qualitative and quantitative analysis was performed on the data derived from the thermal analysis. According to ASTM C1872 [31], the mass percentage of a specific component can be determined under the assumption of component stoichiometry and purity, utilizing the following equation:

$$P = \frac{M \times \Delta_m}{C \times M_0} \times 100$$

where, P accounts for the mass percent of component in the original powder, M is the molar mass of pure stoichiometric component, Δ_m is the mass change within a temperature range, C is the mass loss per mole of pure stoichiometric component due to thermal decomposition, and M_0 is the original mass of sample. In this study, the focus was on the mass of calcium carbonate, with CO₂ being the decomposed component. Molecular masses of 100.08 g/mol for calcium carbonate and 40.08 g/mol for carbon dioxide were utilized in the equation. The approximate percentage uptake of calcium carbonate was then calculated:

$$\text{CaCO}_3 \text{ Uptake \%} = \frac{P}{100 - P} \times 100$$

To examine the impact of carbonation on pore structure, nitrogen adsorption/desorption testing was utilized. Adsorption involves the sticking of gas molecules onto the surface of a solid due to van der Waals attraction forces, while desorption entails the removal of gas molecules from the surface of a solid. Increasing the gas pressure leads to higher adsorption rates, while decreasing pressure results in increased desorption rates. By applying this process to the porous medium and measuring the amount of gas adsorbed/removed across a wide range of relative pressures at a constant temperature, isotherms can be achieved. AAM pulverized paste specimens were dried and degassed in a vacuum oven. The testing temperature was set to 77 K, and the relative pressure range was set from 0.05 to 0.95. The specific surface area (SSA) was then calculated using the Brunauer–Emmett–Teller (BET) theory [32,33].

The mechanical properties of microstructure components were determined using nanoindentation. In the testing, a Hysitron Tri-

boindenter with a Berkovich tip (i.e., pyramidal shape) was utilized. A quasi-static force-controlled mode was employed for the indentation, where a trapezoidal load function was defined with three steps: 10 seconds loading with a rate of 400 $\mu\text{N}/\text{sec}$, 5 seconds holding, and 10 seconds unloading with a rate of 400 $\mu\text{N}/\text{sec}$. The holding period was considered to eliminate any creep effects. The maximum load value (4000 μN) was selected after several preliminary displacement-controlled indents to determine the required loading for the desired indentation depth. The force-displacement data were fitted by applying contact mechanics using the Oliver and Pharr method [34]. The reduced modulus (E_r), which represents the elastic property of both the indenter tip and the testing specimen, is determined as follows:

$$E_r = \frac{1}{\beta} \times \frac{S}{2} \times \frac{\sqrt{\pi}}{\sqrt{A_c}}$$

where, the dimensionless correction factor β accounts for the shape of the indenter tip, which is equal to 1.081 for a Berkovich tip; the parameter S refers to the contact stiffness of the material and is defined as the initial slope of the unloading part of the load–displacement curve at the maximum indentation depth; and A_c is the projected contact area at the peak load (P_{max}). The reduced modulus (E_r) is related to the elastic modulus (E_s) and Poisson's ratio (ν_s) of the tested material. The E_s of the material is then calculated as follows:

$$\frac{1}{E_r} = \frac{1 - \nu_s^2}{E_s} + \frac{1 - \nu_i^2}{E_i}$$

where, E_i and ν_i are the Young's modulus and Poisson's ratio of the diamond indenter tip, which are equal to 1140 GPa and 0.07, respectively. The hardness (H) of each indent is determined as follows:

$$H = \frac{P_{max}}{A_c}$$

3. Results and discussion

3.1. pH measurement of pore solution

The pH value is a critical parameter in the investigation of carbonation in AAM binders due to the direct impact of pore solution pH on several key processes, including calcium leaching, CO_2 dissolution, hydrotalcite formation, and calcium carbonate nucleation and precipitation [35]. Fig. 4 illustrates the results of pH measurements conducted on the pore solutions extracted using the hot water-extraction method. AAM pulverized paste samples were analyzed before and after CO_2 curing. Prior to CO_2 curing (0d), the pH values vary from 11.4 to 12.4,

with the AAS binder exhibiting the highest alkalinity and the AASFM specimen displaying the lowest alkalinity. As expected, higher calcium content in the system correlates with higher alkalinity. AAM binders containing the MgO additive exhibit a lower pH value compared to the binder without the additive. Furthermore, the pH levels of AAM binders exceed the range in which M–S–H gel remains stable [25].

Following the initial cycle of CO_2 curing (28d), a noticeable decrease in pH is observed in all specimens. However, the extent of the pH drop varies among the different binders. The AAS binder, initially having the highest alkalinity, experiences the most significant pH drop (pH = 9.7), while the AASFM binder, with the lowest initial pH, demonstrates the smallest pH drop (pH = 10.5). Consequently, the alkalinity ranking becomes almost inverted after 28 days of CO_2 curing. The pH results further indicate the occurrence of pH drop due to carbonation. The pH results also explain that binders with higher alkalinity are more susceptible to pH drop, emphasizing the governing role of pH levels. The reason behind this can be elaborated upon as the dissolution of CO_2 is significantly greater at elevated pH levels due to the relationship between the amount of dissolved CO_2 and the alkalinity of the solution [36]. Additionally, after the dissolution of CO_2 into the pore solution, speciation of carbonate and bicarbonate ions occurs quickly. At higher pH levels, carbonate ions are more stable than bicarbonate ions, and the higher concentration of carbonate ions favors the precipitation of calcium carbonate [37].

After 56 days of CO_2 curing, no further pH drop is observed in the binders; instead, an increase in pH is noted. However, this pH increment is not significant across the four groups of binders, except for the AAS binder, which rebounds to approximately 11.3. This behavior could be attributed to the dissolution of alkali metals or alkaline earth metals ions (such as sodium or calcium ions) from the solid matrix into the pore solution [35,38].

Last but not the least, in our studies, the pH of the pulverized paste samples did not drop below 9.0. It is important to note that experiments using a phenolphthalein indicator have often employed pH 9.0 as a threshold for measuring carbonation ingress, which can lead to incorrect interpretations. It should be emphasized that pH drop during carbonation is a continuous process which can be misled by a specific threshold.

3.2. X-ray diffraction (XRD) analysis

X-ray diffraction (XRD) was utilized as the analytical technique for chemical analysis, specifically to examine the crystalline phases and polymorphs of calcium carbonate. Fig. 5 presents the XRD patterns of AAM binders before CO_2 curing, collected after regular 28 days of high-moisture curing. All uncarbonated binders exhibit a strong peak at a diffraction angle of 11.45° , indicating the presence of hydrotalcite. A broad hump observed between 25 and 35° can be related to (C, N)–A–S–H gel, where the AAS-C0 binder shows the strongest hump. Furthermore, the AASF-C0 and AASFM-C0 binders exhibit distinct peaks of quartz (SiO_2) and mullite ($3\text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2$). Quartz and mullite are commonly considered as inert phases, where their presence were identified in the raw fly ash XRD analysis (refer to Fig. 1 and Fig. 2). Other minor phases vary among different binders. The AASF-C0 binder exhibits peaks of natron ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$), while the AASM-C0 binder displays peaks of magnesite (MgCO_3). The AASFM-C0 binder exhibits peaks of gaylussite ($\text{Na}_2\text{Ca}(\text{CO}_3)_2 \cdot 5\text{H}_2\text{O}$), identified as a metastable early-stage carbonation product. During later stages of carbonation, gaylussite can dissolve into the pore solution, releasing sodium ions [39]. This dissolution process contributes to the rise in pH of the pore solution as it was observed in the previous section.

AAM binders display several peaks corresponding to calcium carbonate polymorphs. These carbonate phases formed due to natural atmospheric carbonation during the initial curing. The major polymorphs of calcium carbonate are identified as calcite and vaterite. The presence of vaterite is an intriguing observation, given that vaterite is a

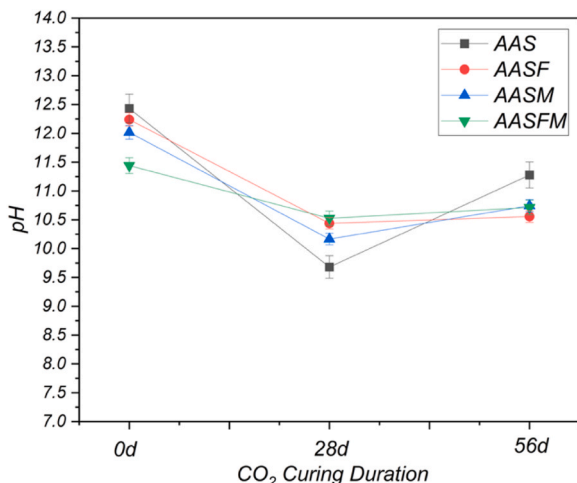


Fig. 4. pH measurements of AAM binders at different stages of 1 % CO_2 curing.

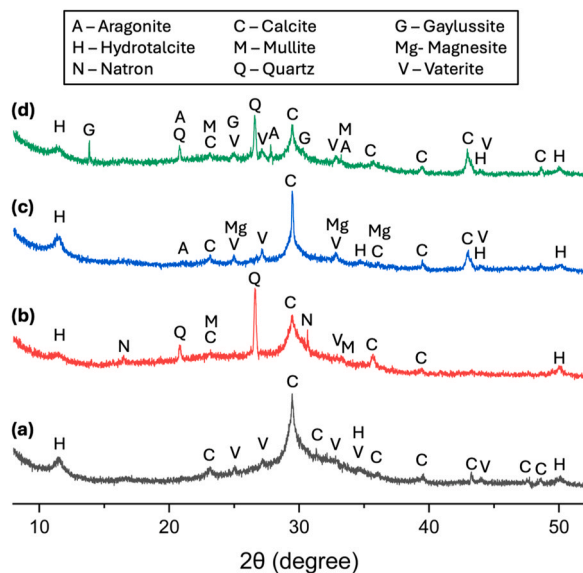


Fig. 5. XRD patterns- uncarbonated binders: (a) AAS-C0, (b) AASF-C0, (c) AASM-C0, and (d) AASFM-C0.

metastable calcium carbonate polymorph that typically undergoes recrystallization into calcite. However, as noted by Zhao et al. [40], hydrated phases on the surface of calcium carbonates could prevent vaterite from converting to calcite.

Fig. 6 and Fig. 7 illustrate the XRD patterns of AAM binders after 28 and 56 days of CO₂ curing. In the AAS and AASF binders, two calcium carbonate polymorphs (i.e., calcite and vaterite) are evident. However, in the AASM and AASFM binders, all three major calcium carbonate polymorphs (i.e., calcite, aragonite, and vaterite) are apparent. It can be implied that a higher MgO content in AAM favors the precipitation of aragonite during carbonation.

Apart from calcium carbonate phases, other metal carbonate crystals are also detected in certain carbonated AAM binders. In the AASF-28C1 and AASF-56C1 binders, trona (Na₃H(CO₃)₂•2H₂O) and natron (Na₂CO₃•10H₂O), both hydrous sodium carbonate crystals, are present. Comparing with the AASF-C0 binder, only natron is found in the AASF-C0 binder. Thermodynamic calculations by Bernal et al. [41], suggested that at room temperature and atmospheric CO₂ concentration, natron

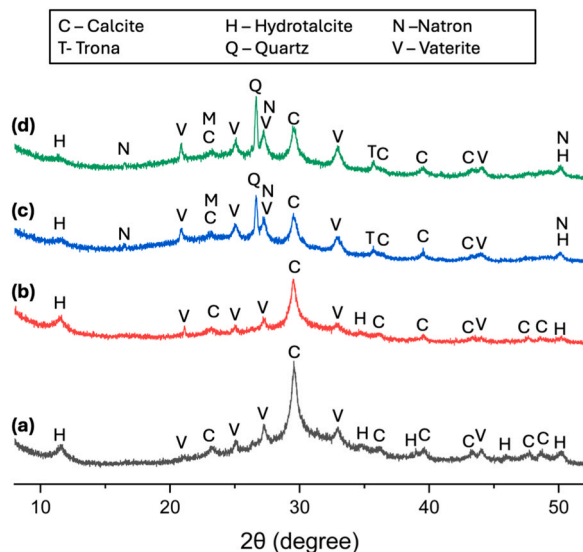


Fig. 6. XRD patterns- AAS and AASF binders after 28 and 56 days of CO₂ curing: (a) AAS-28C1, (b) AAS-56C1, (c) AASF-28C1, and (d) AASF-56C1.

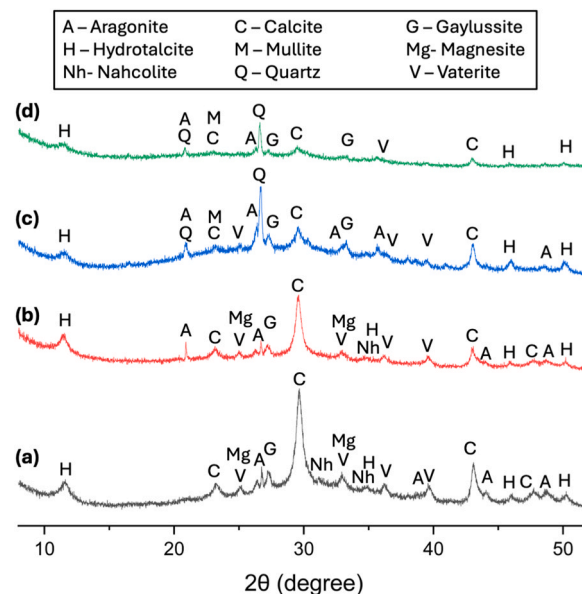


Fig. 7. XRD patterns- AASM and AASFM binders after 28 and 56 days of CO₂ curing: (a) AASM-28C1, (b) AASM-56C1, (c) AASFM-28C1, and (d) AASFM-56C1.

would precipitate. However, a slight increase in CO₂ concentration would favor the precipitation of trona. Further increments in CO₂ concentration would favor the precipitation of nahcolite (NaHCO₃) over natron and trona. In this study, nahcolite peaks are observed in AASM-28C1 and AASM-56C1 binders. The formation of nahcolite instead of natron potentially causes a pore solution with lower pH [42, 43].

Fig. 8 summarizes the quantitative XRD analysis using the Rietveld refinement method. In the AAS binders (Fig. 8a), calcite emerges as the principal calcium carbonate polymorph, accounting for 76–78 % of crystalline phases by weight. Notably, CO₂ curing appears to leave the ratios between crystalline phases largely unaffected, potentially only inducing changes in amorphous or poorly crystalline phases. Further explanation on this observation will be provided in the subsequent thermal analysis section.

In the AASF binders (Fig. 8b), a distinct difference is observed before and after CO₂ curing. In the AASF-C0 binder, calcite and natron dominate as primary phases. Following CO₂ curing, there is a significant decrease in natron weight percentage, accompanied by an increase in trona and vaterite weight percentages, while calcite remains relatively unchanged. As explained earlier, accelerated carbonation could favor the precipitation of trona over natron.

Regarding the AASM binders (Fig. 8c), hydrotalcite emerges as the dominant crystalline phase before CO₂ curing, comprising 44.9 % by weight. Among the calcium carbonate polymorphs, vaterite exhibits a higher weight percentage compared to calcite and aragonite. Additionally, magnesium carbonate precipitates in the form of magnesite, constituting 8.5 % of crystalline phases by weight. After CO₂ curing, there is a decrease in weight percentages for hydrotalcite, vaterite, and magnesite, with concurrent increases observed for calcite, aragonite, and nahcolite phases. The transformation of vaterite into calcite (and aragonite) is noted due to vaterite's metastable nature, which converts to calcite at later stages of carbonation. Notably, the weight percentage of nahcolite exhibits an interesting behavior, with the AASM-28C1 and AASM-56C1 containing 14.1 % and 4.2 %, respectively, suggesting a precipitation-dissolution cycle. Previous studies have suggested that the pH of the pore solution may be related to the weight percentage of nahcolite, with higher nahcolite weight percentages resulting in lower pH levels. Consistent with this hypothesis, the AASM-28C1 exhibits a lower pH than the AASM-56C1.

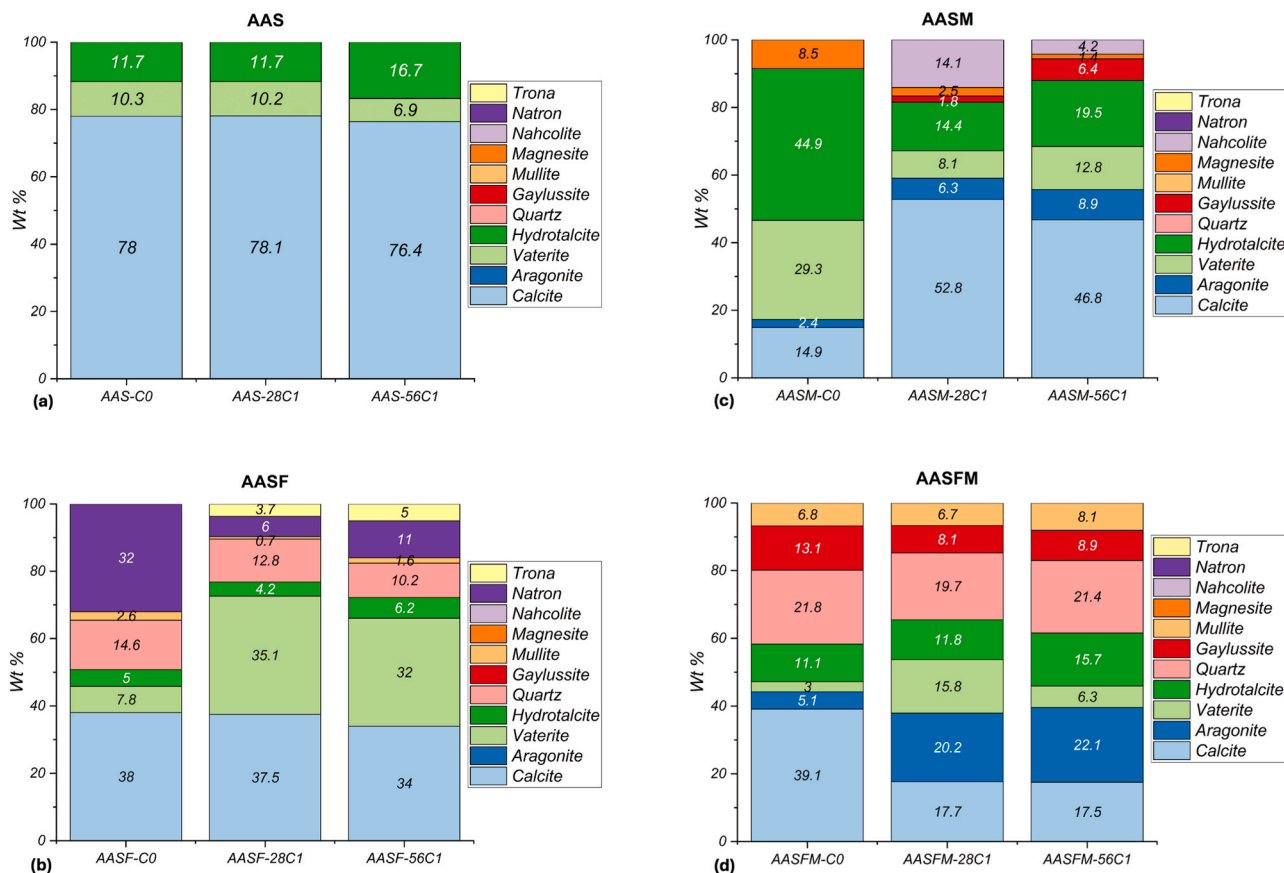


Fig. 8. Quantitative XRD analysis: (a) AAS, (b) AASF, (c) AASM, (d) AASFM.

In the AASFM binders (Fig. 8d), calcite emerges as the primary calcium carbonate polymorph, accompanied by the presence of gaylussite at 17.1 % by weight. Following CO₂ curing, the reactions in the AASFM binders take a different direction compared to the AASM binders. Specifically, there is a decrease in weight percentages for gaylussite and calcite, while weight percentages for vaterite and aragonite increase. Additional explanation to this observation is presented in the following thermal analysis section.

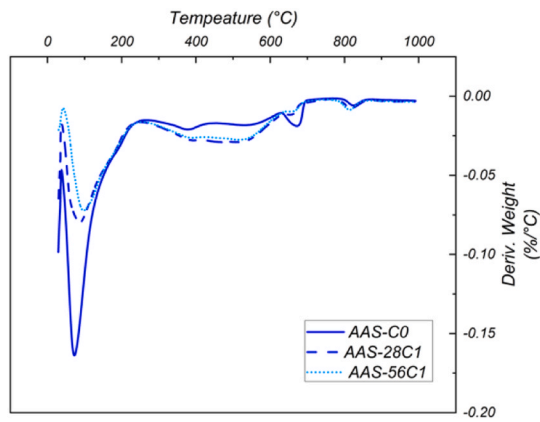
3.3. Thermal analysis

A combined TGA and DSC methodology was employed to thermally analyze the AAM binders before and after carbonation. This methodology allowed for the detection and characterization of phase transitions occurring during the carbonation process. The derivative of the thermogravimetry (DTG) analysis with respect to temperature was used to identify the peaks of weight changes. Fig. 9 and Fig. 10 present the DTG and DSC curves, respectively, for all 12 binders investigated in this study. It is important to note that the heat flow curve is plotted in "Exo Down" format, where an exothermic peak is represented as a downward deflection and an endothermic curve is represented as an upward deflection in the DSC curve.

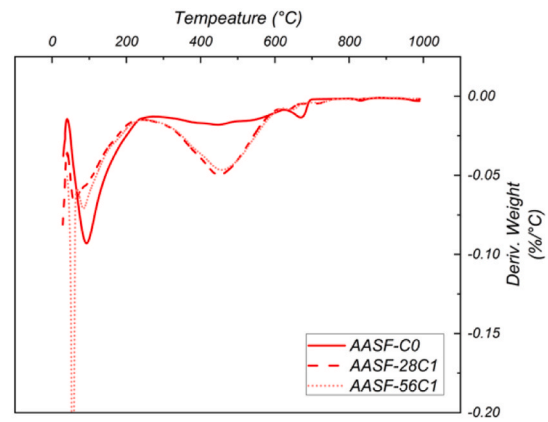
To determine the decomposition temperatures of these minerals, relevant literature and databases were reviewed. Despite some inconsistencies in the literature, approximate temperature ranges for the significant phases were established. The C-A-S-H gel experiences water loss over a broad temperature range (30–600 °C) due to loosely bound and structural water removal [30]. However, the main weight loss of the gel occurs at approximately 50–250 °C [24]. The thermal breakdown of hydrotalcite involves interlayer water elimination, dihydroxylation, [CO₃]²⁻ reduction to CO₂, double layer structure disintegration, and

subsequent recrystallization [30,44]. Parker et al. [10] employed TGA coupled with mass spectrometry and found that the decomposition temperature range of hydrotalcite can be slightly altered based on the carbonate ion content in the interlayer. Hydrotalcite with lower carbonate ion content exhibited a CO₂ release peak at 250–400 °C, while hydrotalcite with higher carbonate ion content showed a peak at 400–450 °C for CO₂ release. Crystalline calcium carbonate typically decomposes into calcium oxide (CaO) and carbon dioxide (CO₂) between 600 and 950 °C. However, the exact positions of decarbonation peaks depend on the fineness of calcium carbonates or presence of hemi- and mono-carbonates [30,31]. On the other hand, amorphous calcium carbonate loses physically-bound and structural water from 50 to 250 °C and decarbonates from 400 to 600 °C [30,45]. It is evident that there is a slight overlap in the decomposition temperatures of hydrotalcite and amorphous calcium carbonate. However, in this study, it is assumed that the range of 250–400 °C corresponds to hydrotalcite, while the range of 400–600 °C corresponds to amorphous calcium carbonate. Thus, four critical temperature ranges have been identified in the thermal analysis of this study: 30–250 °C, 250–400 °C, 400–600 °C, and 600–1000 °C.

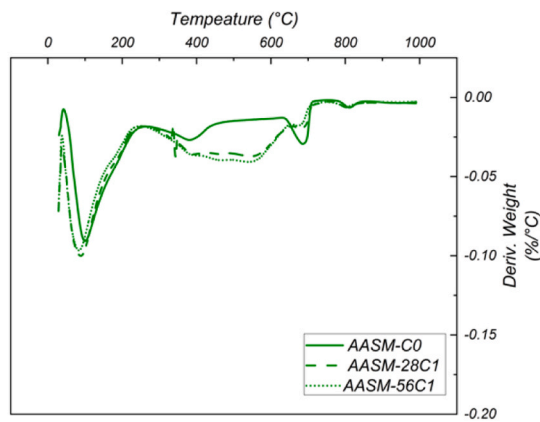
Within the first temperature range, a significant weight loss percentage (WL%) is observed in all binders, and both the DSC and DTG curves exhibit peaks with high intensity. The upward peak in the DSC curve indicates an endothermic process, implying an evaporation event. Within the second critical temperature range, the DTG curves of the CO₂-cured AAS, AASM, and AASFM binders show a visible shoulder, and a low-intensity endothermic peak is observed in the DSC curve. Within the third critical range, differences between the DTG curves of carbonated and uncarbonated binders become evident across all binder types. Furthermore, comparing the DTG curves of the 28-day and 56-day CO₂-cured binders reveals a similar weight change behavior, indicating that carbonation did not precipitate additional products after 28 days of



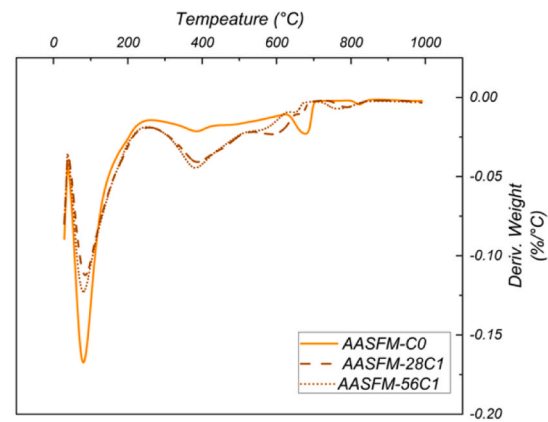
(a) AAS group of binders



(b) AASF group of binders



(c) AASM group of binders



(d) AASFM group of binders

Fig. 9. Comparison of DTG curves at different cycles of CO₂ curing.

curing.

Analyzing the DTG curves between 250 and 600 °C, carbonated AAS and AASM binders exhibit a wide peak that can potentially be deconvoluted into two peaks. The AASF binder displays only a single peak. Although the AASFM binder shows two peaks, the second peak has a lower intensity than the first peak. These observations support the interpretations of the XRD analysis, which presented the presence of hydrotalcite in the AAS and AASM binders and suggested its higher content than in the AASF and AASFM binders. Finally, a general comparison of the DSC curves highlights an interesting finding that the heat flow curve of the 28-day CO₂-cured sample consistently remains lower than the 0-day and 56-day curves throughout the temperature range of 250–600 °C.

A quantitative analysis was performed on the TGA data using a stepwise methodology, and the results are summarized in Table 5. The results show that calcium carbonate was precipitated primarily in amorphous form than crystalline form as the weight loss % between 400 and 600 °C is relatively higher than 600–1000 °C.

Next, the CO₂ mass was used to calculate the calcium carbonate uptake percentage (see Fig. 11). The calcium carbonate uptake before and after 28 days of carbonation reveals a noticeable increase after carbonation. Although the calcium carbonate uptake is growing in all binders, the AASM-28C1 exhibits a higher value (30.60 %) than other cases. After 56 days of CO₂ curing, the trend of the results is different.

Only the AASM-56C1 specimen shows a slight increase in calcium carbonate uptake. The AAS-56C1 and AASF-56C1 specimens exhibit nearly equal uptake compared to the 28-day cured samples. Referring to the quantitative XRD analysis, it was notable that in the AAS binders, the weight percentage of crystalline calcium carbonate phases did not increase following CO₂ curing. However, thermal analysis reveals that both AAS-28C1 and AAS-56C1 binders exhibit a higher uptake of calcium carbonate compared to the AAS-C0 binder. This observation suggests the possible formation and stability of amorphous calcium carbonate during CO₂ curing, thereby hindering further crystallization of calcium carbonates. Another interesting observation pertains to the AASFM binders. The AASFM-56C1 specimen exhibits a lower uptake percentage than the AASFM-28C1 binder. This downward trend can be partially explained by QXRD analysis, wherein the weight percentage of vaterite notably decreased when comparing the AASFM-28C1 and AASFM-56C1 binders. Furthermore, the AASFM-28C1 binder potentially contains a higher content of amorphous calcium carbonate, which, unlike other AAM binders studied, is less stable. The integration of XRD and TGA results implies that, although the AASFM binder includes the MgO additive in its formulation, the stability of amorphous calcium carbonate is not as pronounced as in other AAM binders. Conversely, the weight percentage of hydrotalcite increases after CO₂ curing.

One of the primary objectives of this study was to investigate the effect of MgO on the AAM carbonation mechanism, based on two

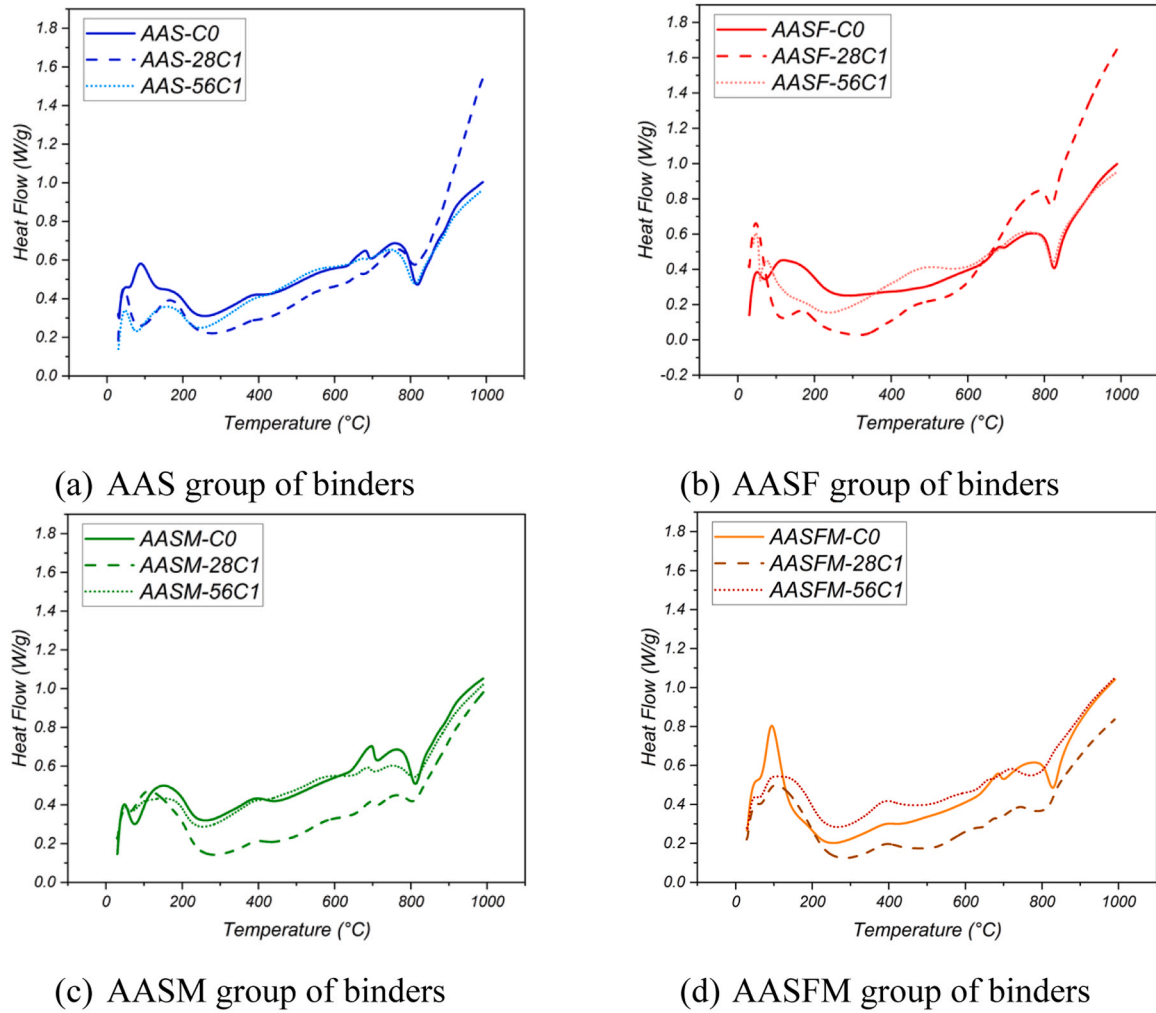


Fig. 10. Comparison of DSC curves at different cycles of CO₂ curing.

Table 5

TGA quantitative analysis.

Binder ID	Weight Loss (%)			
	30–250 °C	250–400 °C	400–600 °C	600–1000 °C
AAS-C0	14.84	2.67	3.48	2.05
AASF-C0	9.91	2.16	3.13	1.46
AASM-C0	10.44	3.33	3.22	2.93
AASFM-C0	17.19	2.67	3.55	2.02
AAS-28C1	10.05	3.3	5.37	2.07
AASF-28C1	8.14	3.51	6.89	1.22
AASM-28C1	12.23	3.87	7.08	3.24
AASFM-28C1	13.09	4.34	5.51	2.14
AAS-56C1	8.62	3.18	5.03	1.97
AASF-56C1	9.85	3.38	6.79	1.21
AASM-56C1	11.34	4.00	7.69	3.00
AASFM-56C1	13.65	4.59	3.25	1.87

underlying hypotheses: 1) the formation of hydrotalcite and 2) the stabilization of amorphous calcium carbonate. This study reveals dissimilar and unique behaviors among AAM binders during carbonation. While all AAM binders exhibited the existence and formation of hydrotalcite, the weight percentage and trends differed. When comparing the AASF and AASFM binders, with the latter possessing a higher MgO/SiO₂ ratio than the former (i.e., 0.15 versus 0.46), observations indicated a heightened formation of hydrotalcite in AASFM binders, yet confirmation of stable amorphous calcium carbonate formation was inconclusive. Similarly,

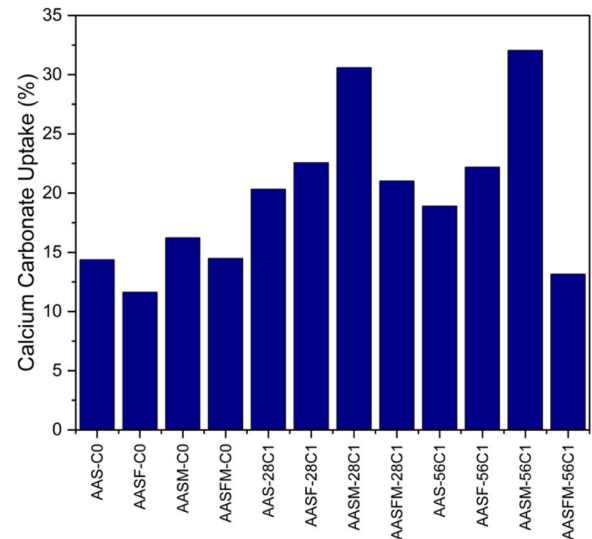


Fig. 11. Quantitative thermal analysis.

comparing AAS and AASM binders, with the latter having a higher MgO/SiO₂ ratio than the former (i.e., 0.30 versus 0.60), revealed a higher hydrotalcite content in AASM binders before CO₂ curing. After CO₂ curing, the amount of hydrotalcite in AASM binders notably decreased,

becoming similar to that in the AAS binders after 56 days of carbonation. Regarding the stability of amorphous calcium carbonate, both AAS and AASM binders contained stable forms, yet in the AASM binders, calcium carbonate crystals could still precipitate, while the AAS binders displayed lower recrystallization.

3.4. Nitrogen adsorption/desorption

Carbonation is a physiochemical process that can alter the pore structure. To investigate the pore structure related to carbonation, the nitrogen adsorption/desorption test was conducted on pulverized AAM pastes before carbonation and after 56 days of carbonation. The isotherms are plotted in Fig. 12, where a significant nitrogen gas uptake can be seen in all binders except AASF-C0 and AASFM-C0 binders. The hysteresis loop of the isotherm curves can be classified as Type IV isotherms, indicating complete pore filling and capillary condensation. The form of the isotherms also suggests that the system is composed of disordered (inkbottle) pores, as the form of the hysteresis loop is close to H2 per original IUPAC classification [46]. Another important observation is regarding the gas uptake, where carbonated AAM binders display a significantly higher gas uptake than uncarbonated ones. This suggests a higher porosity within the 1–100 nm pore diameter range, comprising gel pores and small capillary pores, as proposed by Wang et al. [47] and Yang et al. [48].

The BET specific surface area (SSA) was calculated for different samples and the results are plotted in Fig. 13. SSA increases significantly after carbonation in all cases of AAM binders, which further confirms isotherms observations. For example, SSA of the AAS-C0 and AAS-56C1 binders are 4.8504 and 13.8432 m²/g, respectively. The AASM-56C1 and AASFM-56C1 binders demonstrate the highest SSA which can highlight the role of magnesium in carbonation and CO₂ mineralization. In addition to increased porosity, the growth of SSA can be partly attributed to precipitated calcium carbonate polymorphs. Notably, aragonite demonstrates a relatively higher SSA than calcite and vaterite [49,50]. Earlier XRD findings indicated the presence of precipitated aragonite alongside calcite and vaterite in both AASM and AASFM binders.

3.5. Nanoindentation

Nanoindentation was conducted on AAM pulverized paste samples under three conditions: without CO₂ curing, after 28 days of CO₂ curing, and after 56 days of CO₂ curing. A 20 × 20 grid indentation with a spacing of 5 microns was applied, resulting in 400 data points per sample. The primary objective was to assess the effect of CO₂ curing on

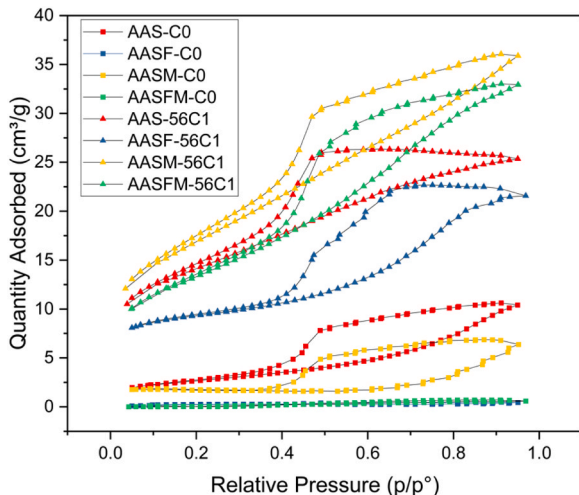


Fig. 12. Nitrogen adsorption/desorption results- isotherms.

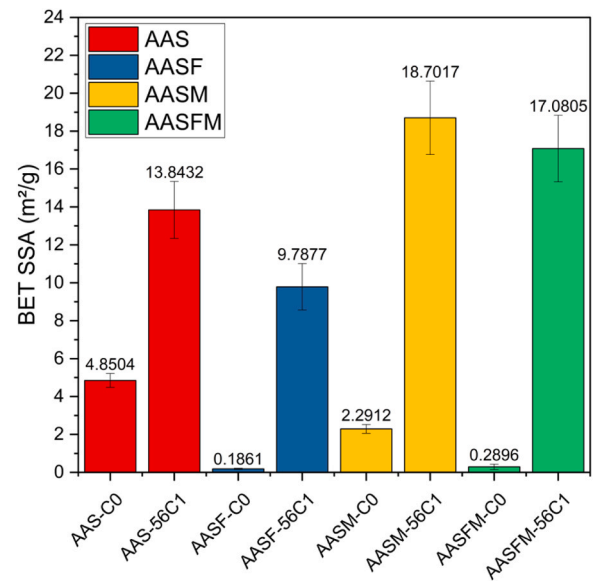


Fig. 13. Nitrogen adsorption/desorption results- BET SSA.

the stiffness of distinct phases: (C, N)–A–S–H and anhydrate phases. The elastic modulus and hardness values were calculated as described in the methodology section.

Given the heterogeneous nature of cementitious materials, the nanoindentation data points were subjected to grouping using the k-means clustering algorithm. This algorithm is well-known for its reliability in clustering data points and determining the cluster center as an average value. To determine the optimal number of clusters, the cubic clustering criterion (CCC) was utilized. The CCC evaluates the sum of squared Euclidean distances between each data point and its assigned cluster center, taking into account the number of clusters and the data's dimensionality. It can effectively identify a clustering solution that maximizes the sum of squared distances between data points and their respective cluster centers.

The results of the CCC analysis are summarized in Table 6, where the k-means clustering algorithm was applied to data points ranging from 2 clusters to 6 clusters, and the CCC values were calculated. It is observed that the k-means clustering with 2 clusters generally yielded the highest CCC value among the tested number of clusters. Thus, 2 total clusters were selected for clustering purposes.

Fig. 14 presents a scatter plot of the nanoindentation data, depicting the relationship between hardness and elastic modulus. The data points were segregated using red and green color ellipses, and the centroid of each cluster was determined. Statistical analysis conducted on the data points within each cluster reveals that the dominant cluster is represented by the red ellipse. The red clusters encompass a larger number of

Table 6
CCC analysis on the number of clusters.

Specimen ID	Number of Clusters				
	2	3	4	5	6
AAS-C0	12.0383	7.0864	6.4926	7.6142	7.2024
AAS-28C1	15.7017	9.2593	8.3148	8.7540	8.0639
AAS-56C1	11.0388	7.2311	4.8215	4.9451	2.4753
AASF-C0	2.6390	-0.0570	1.6940	-0.3030	-2.2570
AASF-28C1	4.2495	-2.4207	-0.2666	-1.0657	1.4500
AASF-56C1	1.5778	-0.0157	-5.4785	0.3946	1.4723
AASM-C0	8.9291	6.4306	5.5846	4.9586	2.7570
AASM-28C1	6.0899	0.1224	-4.1780	2.2000	0.7673
AASM-56C1	7.3885	5.3276	2.0362	3.6650	2.1358
AASFM-C0	4.0150	2.7713	2.1801	0.4219	-0.5005
AASFM-28C1	5.5043	4.6310	0.6549	0.5274	-2.7670
AASFM-56C1	-0.4077	-2.6722	-1.0905	-4.5760	-1.0220

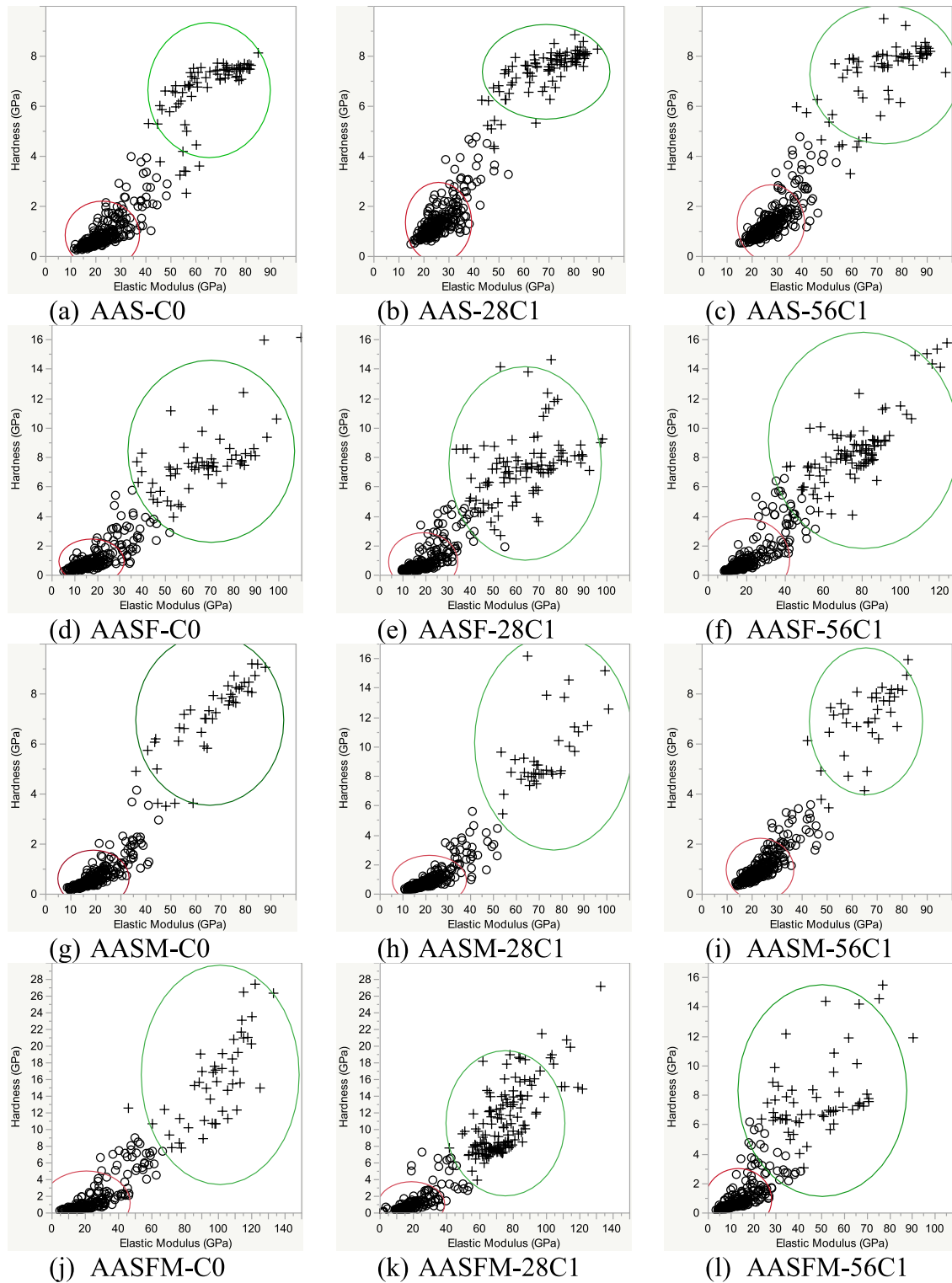


Fig. 14. Nanoindentation results: scattered plots.

data points compared to the green clusters, with the member count ratio ranging from approximately 0.75 to 0.90. The red cluster is likely to represent the properties of the (C, N)–A–S–H gel, while the green cluster corresponds to the properties of anhydrate particles [51]. Given the substantial presence of data points within the (C, N)–A–S–H gel cluster, the subsequent statistical analysis was focused on this phase.

Table 7 provides a summary of the mean and standard deviation values for the elastic modulus and hardness of pozzolanic matrix or (C,

N)–A–S–H gel in various binders.

Initial examination and comparison of the average elastic modulus and hardness values of (C, N)–A–S–H gel prior to CO₂ curing indicate that all four binders exhibit similar properties. When comparing the average elastic modulus and hardness before and after CO₂ curing, an increasing trend can be observed in the AAS, AASF, and AASM binders due to carbonation. Conversely, the AASFM binders display an almost opposite trend, with a decrease in both elastic modulus and hardness.

Table 7

Mean values of (C, N)-A-S-H gel elastic modulus and hardness from clustering centroids.

Binder ID	Elastic Modulus (GPa)	Hardness (GPa)
AAS-C0	22.58 ± 6.93	0.85 ± 0.63
AAS-28C1	25.82 ± 6.17	1.35 ± 0.75
AAS-56C1	27.49 ± 6.28	1.32 ± 0.72
AASF-C0	17.41 ± 8.01	0.93 ± 0.84
AASF-28C1	18.98 ± 7.10	0.91 ± 0.92
AASF-56C1	19.85 ± 10.51	1.09 ± 1.28
AASM-C0	19.03 ± 6.64	0.62 ± 0.53
AASM-28C1	21.74 ± 7.60	0.92 ± 0.80
AASM-56C1	23.14 ± 6.31	0.98 ± 0.58
AASFM-C0	20.61 ± 12.28	1.21 ± 1.78
AASFM-28C1	18.83 ± 9.31	1.05 ± 1.26
AASFM-56C1	13.23 ± 6.83	0.91 ± 0.89

This outcome highlights the diverse effects of carbonation on the micromechanical properties of (C, N)-A-S-H gel. To delve deeper into the impact of carbonation on the elastic modulus and hardness of (C, N)-A-S-H gel in different binders, a statistical analysis is preferred, as discussed further in the subsequent section.

In addition to determining the mean and standard deviations of the elastic modulus and hardness through cluster centroids, k-means clustering is effective in separating the nanoindentation datapoints. The datapoints belonging to (C, N)-A-S-H gel were extracted and treated as an independent distribution. To determine the appropriate method of statistical analysis, such as a T-test, it was necessary to assess whether the distributions followed a normal distribution. The histogram, normal quantile, and outlier box plots for each distribution were thus examined. In addition, the Shapiro-Wilk test was employed to assess the normality of the distributions. All the statistical tests indicated that neither of the distributions follows a normal distribution. Therefore, non-parametric statistical methods were required. The Wilcoxon signed-rank test was thus chosen for the statistical analysis of the nanoindentation datapoints.

The Wilcoxon signed-rank test compares the medians of two paired samples that do not follow a normal distribution [52]. This test involves the null hypothesis (H_0) which assumes that there is no difference between the medians of the paired observations. On the other hand, the alternative hypothesis (H_1) states that there is a significant difference between the medians of the paired observations. The test statistic is based on the ranks of the differences between the paired observations. In addition to p-values, Z values and Hodges-Lehmann estimators were used to test the hypothesis [53].

The results of the Wilcoxon signed-rank test on the elastic modulus and hardness are presented in Table 8 and Table 9, respectively. In Table 8, the null hypothesis is rejected for all the paired tests based on the calculated p-values except two pairs: the pair of AASF-56C1 & AASF-28C1, and the pair of AASFM-C0 & AASFM-28C1. After analyzing the Z and Hodges-Lehmann values, here are the top-ranked results summarized: first, the pair of AAS-C0 and AAS-56C1 shows the most significant

difference in elastic modulus, with the median difference positively skewed towards the AAS-56C1. The pair of AASFM-C0 and AASFM-56C1 shows the next significant difference in elastic modulus with the opposite skewness, where the median difference positively skewed towards the AASFM-C0. Last, the pair of AASM-C0 and AASM-56C1 also shows a significant difference with the median difference positively skewed towards the AASM-56C1.

In Table 9, the null hypothesis is rejected for all paired tests based on the calculated p-values except three pairs: the pair of AAS-56C1 & AAS-28C1, the pair of AASF-56C1 & AASF-28C1, and the pair of AASFM-C0 & AASFM-56C1. Based on the analysis of Z and Hodges-Lehmann values, the top-ranking median differences are the following: the pair of AAS-C0 and AAS-28C1 and the pair of AAS-C0 and AAS-56C1 show the most significant difference in hardness, with the median difference positively skewed towards the AAS-28C1 and AAS-56C1, respectively. Moreover, the pair of AASM-C0 and AASM-56C1 displays a significant difference with the median difference positively skewed towards the AASM-56C1.

In summary, the statistical analyses generally imply a substantial difference in the mechanical characteristics (elastic modulus and hardness) of (C, N)-A-S-H gel of all binders before and after carbonation. However, the extension of carbonation duration from 28 days to 56 days was only statistically significant in a few cases. It should be noted that these findings are based on two distinct curing durations: 28 days and 56 days at 1 % CO₂ concentration, and the effects may vary under different curing conditions.

A comparison between the AAS and AASF binders reveals that the impact of carbonation is more pronounced in AAS binders, leading to greater changes in their micromechanical properties. When comparing the effects of carbonation on the AASM and AASFM binders, a completely different trend can be observed. Directional analysis revealed that CO₂ curing could result in a drop in micromechanical properties of the AASFM binders, while it caused a significant improvement in micromechanical properties of the AASM binders. Comparing nanoindentation results with XRD and TGA suggests a lower amount of calcium carbonate, particularly calcite, in the AASFM binder compared to the AASM binder, partially explaining the differences in micromechanical properties. Additionally, the broad hump in the XRD spectrum between 25 and 35° diffraction angles, associated with (C, N)-A-S-H gel [54], shows less intensity in the AASF and AASFM binders than in the AAS and AASM binders. This indicates a potentially lower quality of (C, N)-A-S-H gel in the AASF and AASFM binders, consequently resulting in lower elastic modulus and hardness.

4. Summary and conclusions

This study examined the effects of CO₂ curing at 1 % concentration on alkali-activated material (AAM) binders incorporated by MgO. Four types of AAM binders were investigated: 100 % slag with/without MgO, 50 % slag-50 % class F fly ash blend with/without MgO, all activated with a powder-type sodium metasilicate. Through a wide range of experimental characterization of the AAM binders before and after

Table 8

Wilcoxon signed-rank test on elastic modulus data.

Level	- Level	Score Mean Difference	Std Err Dif	Z	p-Value	Hodges-Lehmann	Lower CL	Upper CL
AAS-56C1	AAS-28C1	61.187	14.596	4.192	<.0001*	1.770	0.947	2.580
AAS-C0	AAS-28C1	-114.916	14.510	-7.920	<.0001*	-3.488	-4.294	-2.673
AAS-C0	AAS-56C1	-159.035	14.811	-10.738	<.0001*	-5.221	-6.069	-4.389
AASF-56C1	AASF-28C1	-24.883	14.048	-1.771	0.0765	-0.848	-1.746	0.100
AASF-C0	AASF-56C1	-32.007	14.504	-2.207	0.0273	-0.991	-1.944	-0.109
AASF-C0	AASF-28C1	-63.247	14.494	-4.364	<.0001*	-1.840	-2.669	-1.037
AASM-56C1	AASM-28C1	72.599	15.377	4.721	<.0001*	1.85012	1.108	2.597
AASM-C0	AASM-28C1	-101.334	15.232	-6.653	<.0001*	-3.22977	-4.088	-2.350
AASM-C0	AASM-56C1	-148.756	15.408	-9.654	<.0001*	-5.11308	-5.905	-4.287
AASFM-C0	AASFM-56C1	159.508	15.167	10.517	<.0001*	5.073	4.218	5.939
AASFM-C0	AASFM-28C1	13.302	14.325	0.929	0.3531	0.426	-0.491	1.358
AASFM-56C1	AASFM-28C1	-133.677	14.187	-9.423	<.0001*	-4.650	-5.513	-3.790

Table 9

Wilcoxon signed-rank test on hardness data.

Level	- Level	Score Mean Difference	Std Err Dif	Z	p-Value	Hodges-Lehmann	Lower CL	Upper CL
AAS-56C1	AAS-28C1	-6.734	14.596	-0.461	0.6445	-0.016	-0.081	0.050
AAS-C0	AAS-28C1	-189.976	14.510	-13.093	<.0001*	-0.456	-0.514	-0.401
AAS-C0	AAS-56C1	-191.769	14.811	-12.948	<.0001*	-0.447	-0.506	-0.391
AASF-C0	AASF-28C1	63.267	14.494	4.365	<.0001*	0.093	0.054	0.133
AASF-C0	AASF-56C1	55.191	14.504	3.805	0.0001	0.105	0.054	0.152
AASF-56C1	AASF-28C1	-14.122	14.048	-1.005	0.3148	-0.023	-0.063	0.023
AASM-56C1	AASM-28C1	78.713	15.377	5.119	<.0001*	0.124	0.078	0.170
AASM-C0	AASM-28C1	-81.993	15.232	-5.383	<.0001*	-0.115	-0.155	-0.076
AASM-C0	AASM-56C1	-138.947	15.408	-9.018	<.0001*	-0.242	-0.285	-0.198
AASFM-C0	AASFM-56C1	-6.338	15.167	-0.418	0.676	-0.009	-0.048	0.031
AASFM-56C1	AASFM-28C1	-32.918	14.187	-2.320	0.0203	-0.050	-0.097	-0.008
AASFM-C0	AASFM-28C1	-44.707	14.325	-3.121	0.0018	-0.062	-0.098	-0.024

carbonation, the following conclusions and findings were drawn from this study.

- This study indicates that CO₂ curing has a significant impact on pH of AAM binders. A clear trend of pH decrease followed by a partial pH increase was observed. The pH results also illustrated that binders with a higher alkalinity are susceptible to more sensitive pH drop.
- Quantitative XRD analysis was conducted to examine the polymorphs of calcium carbonates. In the AAS and AASF binder categories, both calcite and vaterite polymorphs of calcium carbonate were observed. Conversely, in the AASM and AASFM binder categories, all three major polymorphs of calcium carbonate (calcite, aragonite, and vaterite) were present. This suggests that higher MgO content in AAM binders favored the formation of aragonite during carbonation. AAM binders not only contained calcium carbonates but also exhibited the presence of different types of metal carbonates. In the AASF binder, there was a notable shift before and after CO₂ curing: initially, calcite and natron prevailed, but after curing, natron decreased significantly while trona and vaterite increased. Magnesium carbonate in the form of magnesite was observed in the AASM binder, with nahcolite appearing after CO₂ curing. Finally, in the AASFM binder, calcite was the primary polymorph, accompanied by gaylussite.
- Quantitative thermal analysis performed in this study provided valuable insights into the form and quantity of calcium carbonate formation in AAM binders. Calcium carbonate was precipitated primarily in the amorphous form than the crystalline form. Adding MgO to AAM binders favored the calcium carbonate uptake in AASM binder, while disfavored the calcium carbonate uptake in AASFM binder.
- The coupled quantitative XRD-TGA analysis revealed distinct behaviors among AAM binders during carbonation, with all binders exhibiting hydrotalcite formation, albeit with varying weight percentages and trends. Comparing the AASF and AASFM binders, a higher MgO/SiO₂ ratio led to increased hydrotalcite formation, but stable amorphous calcium carbonate formation could not be observed. Comparing the AAS and AASM binders, the binder with higher MgO/SiO₂ ratio initially showed higher hydrotalcite content, however it decreased significantly after CO₂ curing. Both AAS and AASM binders contained stable amorphous calcium carbonate, but the AAS binders displayed even lower recrystallization.
- The nitrogen adsorption/desorption tests conducted before and after 56 days of carbonation revealed significant changes in pore structure within AAM binders. Carbonated binders exhibited notably higher gas uptake, indicating increased porosity within the 1–100 nm diameter range. This was confirmed by a significant increase in BET specific surface area post-carbonation across all binder types. Particularly, the AASM-56C1 and AASFM-56C1 binders demonstrated the highest surface area.
- Grid nanoindentation coupled with statistical analysis (k-means clustering and Wilcoxon signed-rank test) was carried out on AAM

binders before and after carbonation. It could be concluded that carbonation significantly improved the stiffness of the (C, N)–A–S–H matrix in the AAS, AASF, and AASM binders. However, the AASFM binder showed stiffness loss after carbonation. The comparison between the AAS and AASF binders indicated that carbonation had a more pronounced effect on the AAS binders, resulting in greater changes in their micromechanical properties.

CRedit authorship contribution statement

Shayan Gholami: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jong Suk Jung:** Writing – review & editing, Supervision, Project administration, Investigation, Formal analysis. **Dallas Little:** Writing – review & editing, Supervision, Project administration, Investigation, Conceptualization. **Sukmin Kwon:** Writing – review & editing, Supervision, Project administration, Investigation, Formal analysis. **YONG-RAK KIM:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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